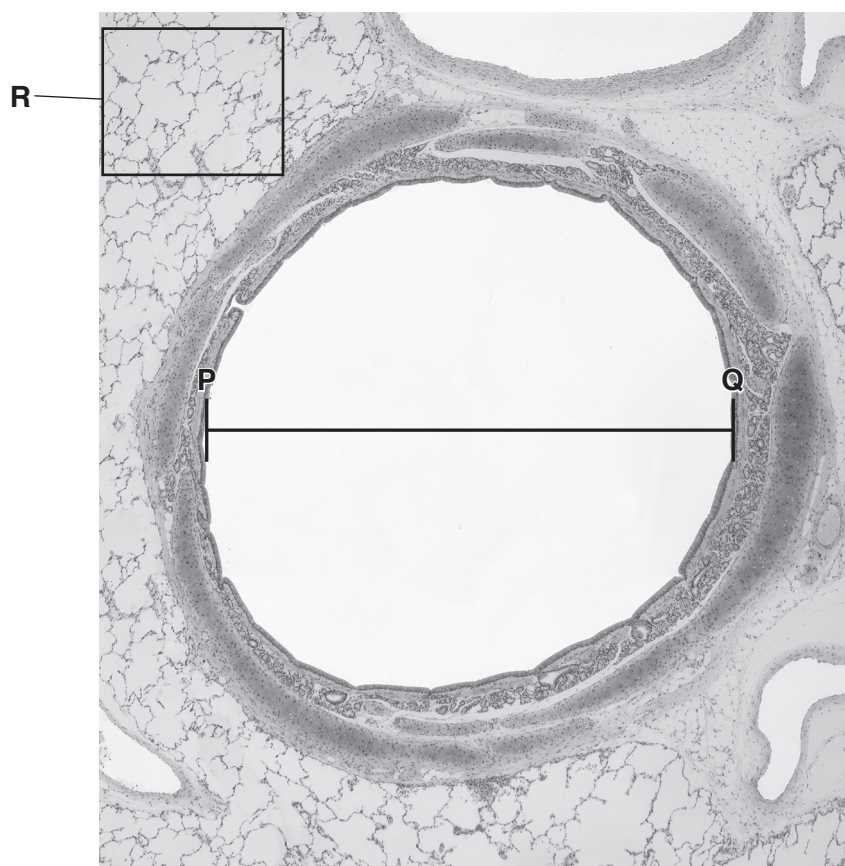


- 2 Fig. 2.1 is a photomicrograph showing a transverse section through a bronchus and the surrounding lung tissue.



magnification x40

Fig. 2.1

- (a) Calculate the actual diameter of the lumen of the bronchus using the line **P–Q**.

Write down the formula used to make your calculation.

Show your working and give your answer to the nearest micrometre (μm).

formula

actual diameter = μm [2]

(b) (i) In Fig. 2.1, the area labelled **R** shows a section through some alveoli.

Name the tissue that lines the air spaces of the alveoli.

..... [1]

(ii) Suggest how the alveoli are adapted for gas exchange.

.....
.....
.....
.....
..... [2]

(c) In the leaves of plants, both gas exchange and transpiration occur through open stomata.

Suggest how the arrangement of cells in the leaf of a dicotyledonous plant contributes to the loss of water by transpiration.

.....
.....
.....
.....
.....
..... [3]

(d) Fig. 2.2 shows the mean transpiration rate of a xerophyte between 08:00 and 19:00.

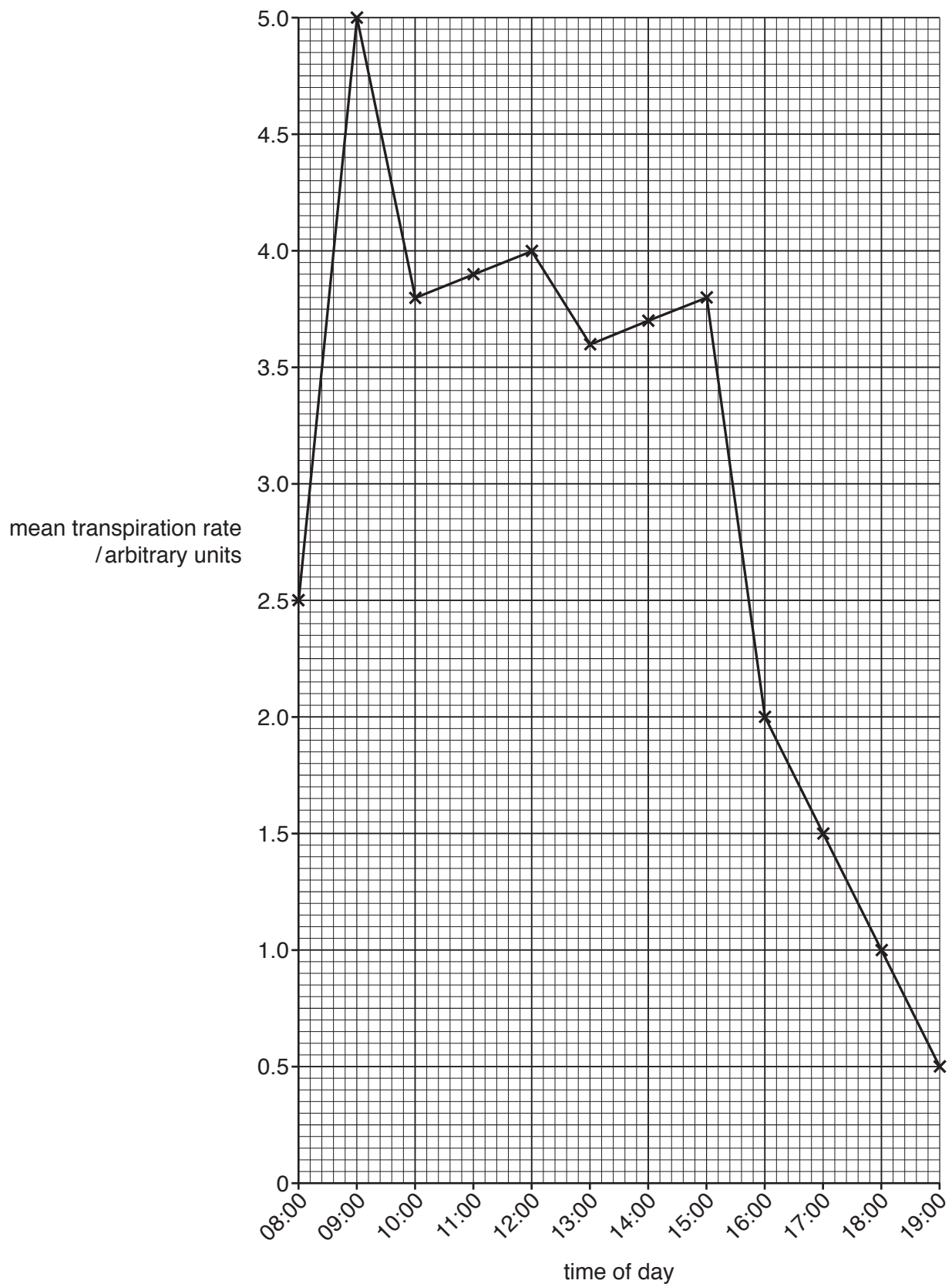


Fig. 2.2

(i) Describe the data shown in Fig. 2.2.

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.....

.....

.....

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.....

.....

..... [3]

(ii) Suggest why the transpiration rate changed between 09:00 and 10:00.

.....

.....

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.....

.....

..... [2]

[Total: 13]